ECE 332: Engineering Electromagnetics II

Lecture 6

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Quick Review

- We began to discuss **electromagnetic (EM) waves**, which occur due to the coupling of the electric and magnetic fields in Maxwell's equations.
- An **unbounded wave** travels through a homogeneous medium unimpeded, without encountering any boundaries.
- Media can be characterized as either **lossless** or **lossy**.
- In a **lossless medium**, $\sigma = 0$, EM waves are never attenuated, and no energy is dissipated.
- In a **lossy medium**, $\sigma \neq 0$, EM waves are attenuated, and energy is dissipated as heat.
- Sources of EM waves generally emit waves with **spherical wavefronts**.

Quick Review (Continued)

- **"Far" from the source**, the wavefronts look **planar**.
- In a **uniform plane wave**, the fields are uniform on parallel planes.
- Our initial discussion will be about **unbounded plane waves**.
- One way that EM waves can be bounded is when they're **guided**, which means that they **move along material structures**.

  Examples

  - **The confinement of radio waves in the HF band** between the ground and the ionosphere.
  - **EM waves carried on a transmission line**, which also highlights the role of coupling between the electric and magnetic fields in the production of EM waves.

Quick Review (Continued)

- The **phasor form of Maxwell's equations** is

$$\nabla \cdot \tilde{\mathbf{E}} = \tilde{\rho}_v / \epsilon \tag{1}$$

$$\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}} \tag{2}$$

$$\nabla \cdot \tilde{\mathbf{H}} = 0 \tag{3}$$

$$\nabla \times \tilde{\mathbf{H}} = j\omega\epsilon_c\tilde{\mathbf{E}}, \tag{4}$$

where

$$\epsilon_c = \epsilon - j(\sigma/\omega) = \epsilon' - j\epsilon'' \tag{5}$$

is the **complex permittivity** and $\tilde{\mathbf{J}} = \sigma\tilde{\mathbf{E}}$. These describe the **spatial behavior** of the **time-harmonic fields**

$$\mathbf{E}(x, y, z, t) = \mathrm{Re}\left\{\tilde{\mathbf{E}}(x, y, z)e^{j\omega t}\right\}, \tag{6}$$

$$\mathbf{H}(x, y, z, t) = \mathrm{Re}\left\{\tilde{\mathbf{H}}(x, y, z)e^{j\omega t}\right\}, \tag{7}$$

$$\rho_v(x, y, z, t) = \mathrm{Re}\left\{\tilde{\rho}_v(x, y, z)e^{j\omega t}\right\}. \tag{8}$$

Quick Review (Continued)

- The phasor form of Maxwell's equations in a **source-free medium** (where $\tilde{\rho_v} = \rho_v = 0$) is

$$\nabla \cdot \tilde{\mathbf{E}} = 0 \tag{9}$$

$$\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}} \tag{10}$$

$$\nabla \cdot \tilde{\mathbf{H}} = 0 \tag{11}$$

$$\nabla \times \tilde{\mathbf{H}} = j\omega\epsilon_c\tilde{\mathbf{E}}. \tag{12}$$

- In a **source-free medium**, the phasors of the electric and magnetic field intensities satisfy the homogeneous wave equations

$$\nabla^2\tilde{\mathbf{E}} - \gamma^2\tilde{\mathbf{E}} = \mathbf{0} \tag{13}$$

$$\nabla^2\tilde{\mathbf{H}} - \gamma^2\tilde{\mathbf{H}} = \mathbf{0}, \tag{14}$$

where γ is the **propagation constant**, defined by

$$\gamma^2 = -\omega^2\mu\epsilon_c. \tag{15}$$

$\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ obey the same equations and will have the same form.

Quick Review (Continued)

- In a **lossless and source-free medium**, $\sigma = 0$, $\epsilon_c = \epsilon$, and the homogeneous wave equation for $\tilde{\mathbf{E}}$ is

$$\nabla^2 \tilde{\mathbf{E}} + k^2 \tilde{\mathbf{E}} = \mathbf{0}, \tag{16}$$

where the **wavenumber** is defined as

$$k = \omega \sqrt{\mu \epsilon}. \tag{17}$$

- In Cartesian coordinates,

$$\tilde{\mathbf{E}} = \tilde{E}_x \hat{\mathbf{x}} + \tilde{E}_y \hat{\mathbf{y}} + \tilde{E}_z \hat{\mathbf{z}} \tag{18}$$

and each component of $\tilde{\mathbf{E}}$ must satisfy the wave equation

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k^2 \right) \tilde{E}_* = 0, \tag{19}$$

where the asterisk can be replaced by x, y, or z.

Quick Review (Continued)

- A **uniform plane EM wave** has electric and magnetic fields that have **uniform properties on an infinite plane**. If the medium is **lossless** and **source-free** and (arbitrarily) **the plane is the** x-y **plane**, this means

$$\frac{\partial \tilde{E}_x}{\partial x} = \frac{\partial \tilde{E}_x}{\partial y} = 0, \quad \frac{\partial \tilde{E}_y}{\partial x} = \frac{\partial \tilde{E}_y}{\partial y} = 0 \tag{20}$$

$$\frac{\partial \tilde{H}_x}{\partial x} = \frac{\partial \tilde{H}_x}{\partial y} = 0, \quad \frac{\partial \tilde{H}_y}{\partial x} = \frac{\partial \tilde{H}_y}{\partial y} = 0 \tag{21}$$

and the **wave equations** for $\tilde{E}_x$, $\tilde{E}_y$, $\tilde{H}_x$, $\tilde{H}_y$ are

$$\frac{d^2 \tilde{E}_x}{dz^2} + k^2 \tilde{E}_x = 0, \quad \frac{d^2 \tilde{E}_y}{dz^2} + k^2 \tilde{E}_y = 0, \tag{22}$$

$$\frac{d^2 \tilde{H}_x}{dz^2} + k^2 \tilde{H}_x = 0, \quad \frac{d^2 \tilde{H}_y}{dz^2} + k^2 \tilde{H}_y = 0, \tag{23}$$

where $\tilde{\mathbf{H}} = \tilde{H}_x \hat{\mathbf{x}} + \tilde{H}_y \hat{\mathbf{y}} + \tilde{H}_z \hat{\mathbf{z}}$.

Quick Review (Continued)

- Under these circumstances, the z components of Ampère's law $\nabla \times \tilde{\mathbf{H}} = j\omega\epsilon\tilde{\mathbf{E}}$ and Faraday's law $\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}}$ imply

$$\tilde{E}_z = \tilde{H}_z = 0, \tag{24}$$

which means **there are no electric or magnetic field components in the direction of propagation**.

- The **general solution** to

$$\frac{d^2\tilde{E}_x}{dz^2} + k^2\tilde{E}_x = 0 \tag{25}$$

is

$$\tilde{E}_x(z) = \tilde{E}_x^+(z) + \tilde{E}_x^-(z) = E_{x0}^+ e^{-jkz} + E_{x0}^- e^{jkz}, \tag{26}$$

where $\tilde{E}_x^+(z)$ is a wave with amplitude E_{x0}^+ in the $+z$ direction and $\tilde{E}_x^-(z)$ is a wave with amplitude E_{x0}^- in the $-z$ direction.

Quick Review (Continued)

- The **phasor of the electric field intensity** for a **uniform plane wave** in the z direction with uniform properties on the x-y plane in a **lossless, source-free medium** has the form

$$\tilde{E}_x(z) = \tilde{E}_x^+(z) + \tilde{E}_x^-(z) = E_{x0}^+ e^{-jkz} + E_{x0}^- e^{jkz}, \tag{27}$$

where $\tilde{E}_x^+(z)$ is a wave with amplitude E_{x0}^+ in the $+z$ direction and $\tilde{E}_x^-(z)$ is a wave with amplitude E_{x0}^- in the $-z$ direction.

- Assuming $\tilde{E}_y = 0$ and $E_{x0}^- = 0$, the **phasor of the electric field** is

$$\tilde{\mathbf{E}}(z) = E_{x0}^+ e^{-jkz} \hat{\mathbf{x}} \tag{28}$$

and the **phasor of the magnetic field** is

$$\tilde{\mathbf{H}}(z) = H_{y0}^+ e^{-jkz} \hat{\mathbf{y}} = \frac{E_{x0}^+}{\eta} e^{-jkz} \hat{\mathbf{y}}, \tag{29}$$

where

$$\eta = \frac{\omega\mu}{k} = \sqrt{\frac{\mu}{\epsilon}} (\Omega). \tag{30}$$

Quick Review (Continued)

- The uniform plane wave is a **transverse electromagnetic wave (TEM)** since $\tilde{\mathbf{E}}$, $\tilde{\mathbf{H}}$, and the propagation direction are perpendicular.
- **Electric and magnetic field intensities** of the uniform plane wave:

$$\mathbf{E}(z,t) = \mathrm{Re}\left\{\tilde{\mathbf{E}}(z)e^{j\omega t}\right\} = |E_{x0}^{+}|\cos(\omega t - kz + \phi^{+})\hat{\mathbf{x}} \quad (\mathrm{V/m}) \tag{31}$$

$$\mathbf{H}(z,t) = \mathrm{Re}\left\{\tilde{\mathbf{H}}(z)e^{j\omega t}\right\} = \frac{|E_{x0}^{+}|}{\eta}\cos(\omega t - kz + \phi^{+})\hat{\mathbf{y}} \quad (\mathrm{A/m}), \tag{32}$$

where

$$E_{x0}^{+} = |E_{x0}^{+}|e^{j\phi^{+}}. \tag{33}$$

These fields are always **in phase** for **lossless media**.

- The **phase velocity** (speed of propagation) of the wave is

$$u_p = \frac{\omega}{k} = \frac{1}{\sqrt{\mu\epsilon}} \quad (\mathrm{m/s}), \tag{34}$$

i.e. the speed of light in the medium.

Quick Review (Continued)

- The **wavelength** of the wave is

$$\lambda = \frac{2\pi}{k} = \frac{u_p}{f} \quad \text{(m)}, \qquad (35)$$

where $f = \omega/2\pi$ is the **frequency**.

- The **intrinsic impedance of free space** is

$$\eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} = 377 \quad (\Omega)$$
$$\approx 120\pi \quad (\Omega). \qquad (36)$$

- We explored the example of an EM plane wave in air.

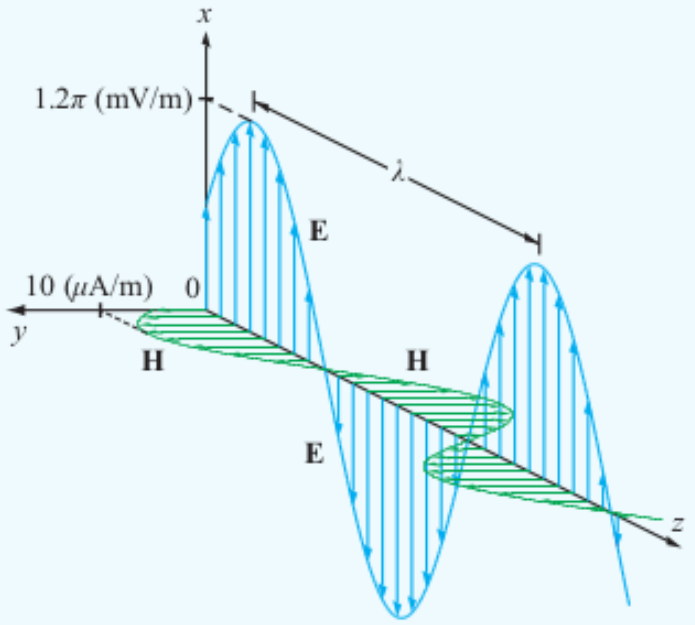

Figure 7-5 Spatial variations of **E** and **H** at $t = 0$ for the plane wave of Example 7-1.

Quick Review (Continued)

- For a **uniform plane wave traveling in direction $\hat{\mathbf{k}}$ in any medium**

$$\tilde{\mathbf{H}} = \frac{1}{\eta}\hat{\mathbf{k}} \times \tilde{\mathbf{E}} \tag{37}$$

$$\tilde{\mathbf{E}} = -\eta\hat{\mathbf{k}} \times \tilde{\mathbf{H}}. \tag{38}$$

- The **directions of E, H, and $\hat{\mathbf{k}}$** are specified by a **right-hand rule** in which curling the fingers of the right hand from the direction of **E** to the direction of **H** causes the right thumb to point in the $\hat{\mathbf{k}}$ direction.

Examples:

- **Wave Propagating Along $+z$ ($\hat{\mathbf{k}} = \hat{\mathbf{z}}$) with E in $\hat{\mathbf{x}}$ Direction**

$$\tilde{\mathbf{E}} = E_{x0}^{+} e^{-jkz}\hat{\mathbf{x}} \tag{39}$$

$$\tilde{\mathbf{H}} = \frac{E_{x0}^{+}}{\eta} e^{-jkz}\hat{\mathbf{y}} \tag{40}$$

- **Wave Propagating Along $-z$ ($\hat{\mathbf{k}} = -\hat{\mathbf{z}}$) with E in $\hat{\mathbf{x}}$ Direction**

$$\tilde{\mathbf{E}} = E_{x0}^{-} e^{jkz}\hat{\mathbf{x}} \tag{41}$$

$$\tilde{\mathbf{H}} = -\frac{E_{x0}^{-}}{\eta} e^{jkz}\hat{\mathbf{y}} \tag{42}$$

Quick Review (Continued)

- **Wave Propagating Along $+z$ ($\hat{\mathbf{k}} = \hat{\mathbf{z}}$) with E in $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ Directions**

$$\tilde{\mathbf{E}} = E_{x0}^{+} e^{-jkz} \hat{\mathbf{x}} + E_{y0}^{+} e^{-jkz} \hat{\mathbf{y}} \quad (43)$$

$$\tilde{\mathbf{H}} = -\frac{E_{y0}^{+}}{\eta} e^{-jkz} \hat{\mathbf{x}} + \frac{E_{x0}^{+}}{\eta} e^{-jkz} \hat{\mathbf{y}} \quad (44)$$

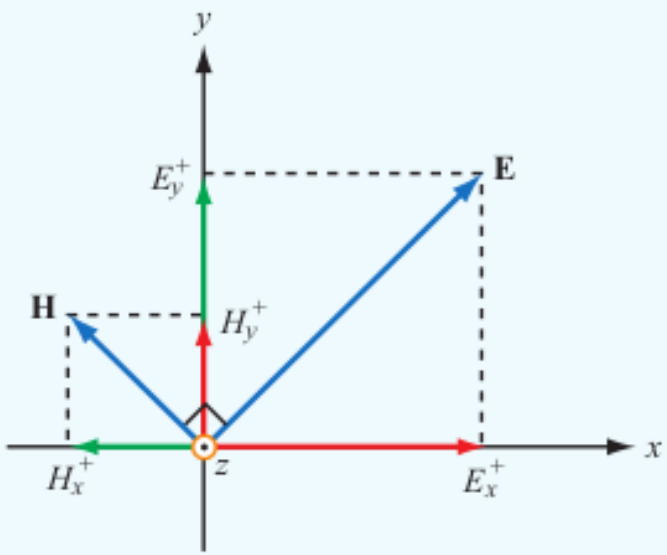

Figure 7-6 The wave (**E**,**H**) is equivalent to the sum of two waves, one with fields (E_x^+, H_y^+) and another with (E_y^+, H_x^+), with both traveling in the $+z$ direction.

Wave Polarization

The **polarization** of a uniform plane wave describes the **curve that the electric field intensity vector E traces over time** in a plane perpendicular to the direction of propagation $\hat{\mathbf{k}}$ at a fixed value of z.

In general, the phasor of the electric field intensity of a uniform plane wave that propagates in the $+z$ direction has the form

$$\tilde{\mathbf{E}}(z) = \tilde{E}_x(z)\hat{\mathbf{x}} + \tilde{E}_y(z)\hat{\mathbf{y}}, \tag{45}$$

where

$$\tilde{E}_x(z) = E_{x0}e^{-jkz} \quad \text{and} \quad \tilde{E}_y(z) = E_{y0}e^{-jkz}. \tag{46}$$

The amplitudes E_{x0} and E_{y0} are **complex-valued** and may be written in **polar form** in terms of their **magnitudes** and **phases**. Since only the phase difference between E_{x0} and E_{y0} and not the values of the phases of E_{x0} and E_{y0} is important to the polarization, it's okay to specify that the phase of E_{x0} as 0 and the phase of E_{y0} as δ. (This is equivalent to setting the phase corresponding to the x-axis to 0.)

Wave Polarization (Continued)

This means defining

$$E_{x0} = a_x \quad \text{and} \quad E_{y0} = a_y e^{j\delta}, \tag{47}$$

where

$$a_x = |E_{x0}| \quad \text{and} \quad a_y = |E_{y0}|. \tag{48}$$

With these definitions, the phasor of $\mathbf{E}$ is

$$\tilde{\mathbf{E}}(z) = (a_x \hat{\mathbf{x}} + a_y e^{j\delta} \hat{\mathbf{y}}) e^{-jkz} \tag{49}$$

and the electric field intensity is

$$\begin{aligned}
\mathbf{E}(z,t) &= \mathrm{Re}\left\{\tilde{\mathbf{E}}(z) e^{j\omega t}\right\} = \mathrm{Re}\left\{(a_x \hat{\mathbf{x}} + a_y e^{j\delta} \hat{\mathbf{y}}) e^{-jkz} e^{j\omega t}\right\} \\
&= \mathrm{Re}\left\{(a_x e^{j(\omega t - kz)} \hat{\mathbf{x}} + a_y e^{j(\omega t - kz + \delta)} \hat{\mathbf{y}})\right\} \\
&= a_x \cos(\omega t - kz) \hat{\mathbf{x}} + a_y \cos(\omega t - kz + \delta) \hat{\mathbf{y}} \\
&= E_x(z,t) \hat{\mathbf{x}} + E_y(z,t) \hat{\mathbf{y}},
\end{aligned} \tag{50}$$

Wave Polarization (Continued)

where

$$E_x(z,t) = a_x \cos(\omega t - kz) \quad \text{and} \quad E_y(z,t) = a_y \cos(\omega t - kz + \delta). \tag{51}$$

The **magnitude** of the electric field intensity is

$$\begin{aligned} |\mathbf{E}(z,t)| &= \sqrt{E_x^2(z,t) + E_y^2(z,t)} \\ &= (a_x^2 \cos^2(\omega t - kz) + a_y^2 \cos^2(\omega t - kz + \delta))^{1/2} \end{aligned} \tag{52}$$

and the **angle** that it makes with the positive x axis is

$$\psi(z,t) = \tan^{-1}\left(\frac{E_y(z,t)}{E_x(z,t)}\right). \tag{53}$$

This angle is called the **inclination angle**.

In general, $\mathbf{E}(z,t)$ will follow an **ellipse** in the x-y plane. However, the ellipse will be a **circle** in some cases. In this case, the polarization is called **circular**. The ellipse can also collapse into a **line segment**. In this case, the polarization is said to be **linear**.

Wave Polarization (Continued)

Linear Polarization

The ellipse is a line segment when $E_x(z,t)$ and $E_y(z,t)$ are **in phase**, which means $\delta = 0$, or **out of phase**, which means that $\delta = \pi$.

When $E_x(z,t)$ and $E_y(z,t)$ are in phase,

$$\mathbf{E}(z,t) = (a_x\hat{\mathbf{x}} + a_y\hat{\mathbf{y}})\cos(\omega t - kz). \tag{54}$$

In this case, the inclination angle is

$$\psi = \tan^{-1}\left(\frac{a_y\cos(\omega t - kz)}{a_x\cos(\omega t - kz)}\right) = \tan^{-1}\left(\frac{a_y}{a_x}\right). \tag{55}$$

This is the angle that the line segment in the x-y plane makes relative to the positive x axis.

Wave Polarization (Continued)

When $E_x(z,t)$ and $E_y(z,t)$ are out of phase,

$$\mathbf{E}(z,t) = (a_x\hat{\mathbf{x}} - a_y\hat{\mathbf{y}})\cos(\omega t - kz). \quad (56)$$

In this case, the inclination angle is

$$\psi = \tan^{-1}\left(\frac{-a_y\cos(\omega t - kz)}{a_x\cos(\omega t - kz)}\right) = \tan^{-1}\left(-\frac{a_y}{a_x}\right). \quad (57)$$

In both cases,

$$|\mathbf{E}(z,t)| = \sqrt{a_x^2 + a_y^2}\,|\cos(\omega t - kz)|. \quad (58)$$

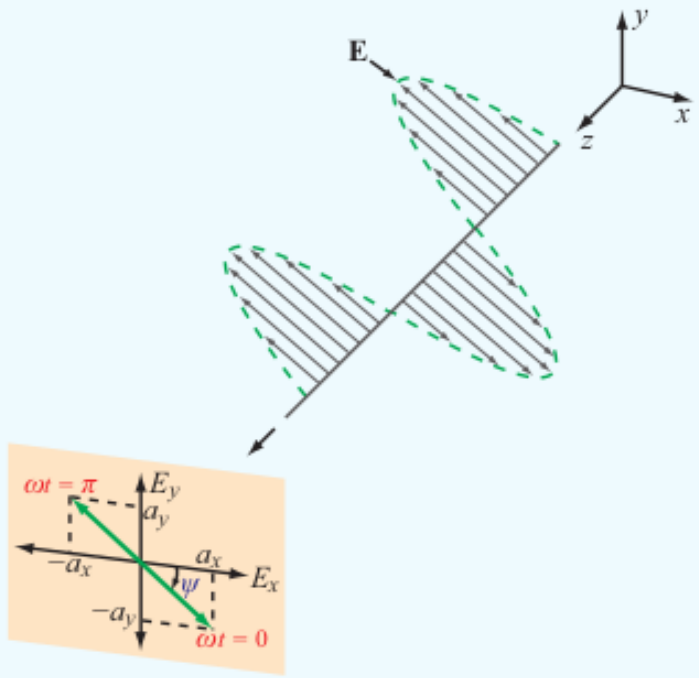

Figure 7-7 Linearly polarized wave traveling in the $+z$ direction (out of the page).

If $a_y = 0$, $\psi = 0$ or π and the wave is **x-polarized**.
If $a_x = 0$, $\psi = \pi/2$ or $-\pi/2$ and the wave is **y-polarized**.

Wave Polarization (Continued)

Circular Polarization

If $a_x = a_y = a > 0$ and $\delta = \pm\pi/2$, then the polarization is **circular**. In this case, there are two possibilities.

- **Left-Hand Circular (LHC) Polarization**

  In this case, $\delta = \pi/2$, which means

$$\tilde{\mathbf{E}}(z) = (a\hat{\mathbf{x}} + ae^{j\pi/2}\hat{\mathbf{y}})e^{-jkz} = a(\hat{\mathbf{x}} + j\hat{\mathbf{y}})e^{-jkz} \tag{59}$$

  and

$$\begin{aligned}\mathbf{E}(z,t) &= \mathrm{Re}\left\{\tilde{\mathbf{E}}(z)e^{j\omega t}\right\} = \mathrm{Re}\left\{a(\hat{\mathbf{x}} + j\hat{\mathbf{y}})e^{-jkz}e^{j\omega t}\right\} \\ &= a\,\mathrm{Re}\left\{e^{j(\omega t - kz)}\hat{\mathbf{x}} + je^{j(\omega t - kz)}\hat{\mathbf{y}}\right\} \\ &= a(\cos(\omega t - kz)\hat{\mathbf{x}} - \sin(\omega t - kz)\hat{\mathbf{y}}).\end{aligned} \tag{60}$$

  So,

$$|\mathbf{E}(z,t)| = \sqrt{a^2(\cos^2(\omega t - kz) + \sin^2(\omega t - kz))} = a \tag{61}$$

Wave Polarization (Continued)

and

$$\psi(z,t) = \tan^{-1}\left(\frac{E_y(z,t)}{E_x(z,t)}\right) = \tan^{-1}\left(\frac{-a\sin(\omega t - kz)}{a\cos(\omega t - kz)}\right)$$
$$= \tan^{-1}(-\tan(\omega t - kz)) = \tan^{-1}(\tan(-(\omega t - kz)))$$
$$= -(\omega t - kz). \tag{62}$$

Since

$$\frac{\partial \psi}{\partial t} = -\omega < 0 \tag{63}$$

(assuming $\omega > 0$), the inclination angle moves in the clockwise direction at a constant rate as t increases. The wave is ***left-hand circularly polarized*** because pointing the left thumb in the direction of propagation causes the left fingers to curl in the direction that **E** moves as time increases.

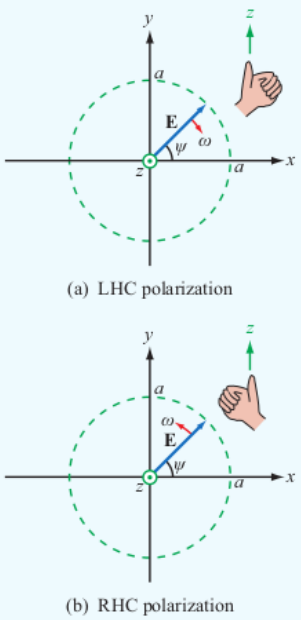

Figure 7-8 Circularly polarized plane waves propagating in the $+z$ direction (out of the page).

Wave Polarization (Continued)

- **Right-Hand Circular (RHC) Polarization**

  Here, $\delta = -\pi/2$,

$$\tilde{\mathbf{E}}(z) = a(\hat{\mathbf{x}} - j\hat{\mathbf{y}})e^{-jkz}, \tag{64}$$

$$\mathbf{E}(z,t) = a(\cos(\omega t - kz)\hat{\mathbf{x}} + \sin(\omega t - kz)\hat{\mathbf{y}}), \tag{65}$$

$$|\mathbf{E}(z,t)| = a, \text{ and } \psi(z,t) = \omega t - kz. \tag{66}$$

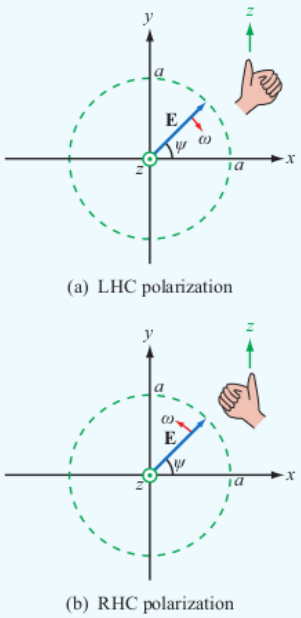

Figure 7-8 Circularly polarized plane waves propagating in the $+z$ direction (out of the page).

Note that for both types of circular polarization, the electric field intensity vector **E** rotates in the opposite sense as a function of z for fixed t as it does as a function of t in a plane perpendicular to the direction of propagation for fixed z.

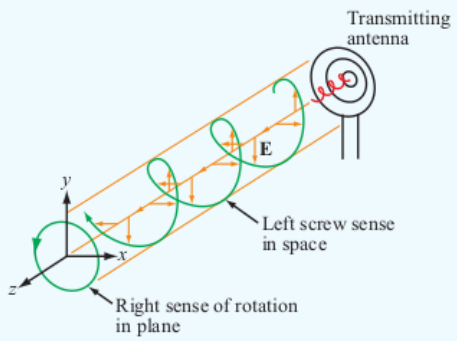

Figure 7-9 Right-hand circularly polarized wave radiated by a helical antenna.

Wave Polarization (Continued)

Example: If a RHC-polarized wave with frequency 100 MHz and electric field magnitude 3 mV/m is traveling in the $+y$ direction in a dielectric medium with $\epsilon = 4\epsilon_0$, $\mu = \mu_0$, and $\sigma = 0$, find $\mathbf{E}(y,t)$ and $\mathbf{H}(y,t)$.

Here,

$$a = |\mathbf{E}(y,t)| = 3 \text{ mV/m}, \tag{67}$$

$$f = 100 \text{ MHz} = 100 \times 10^6 \text{ Hz} = 10^8 \text{ Hz}. \tag{68}$$

So,

$$\omega = 2\pi f = 2\pi \times 10^8 \text{ rad/s} \tag{69}$$

and

$$k = \omega\sqrt{\mu\epsilon} = \omega\sqrt{\mu_0(4\epsilon_0)} = \frac{2\omega}{c} = \frac{4\pi \times 10^8 \text{ rad/s}}{3 \times 10^8 \text{ m/s}} = \frac{4\pi}{3} \text{ rad/m}. \tag{70}$$

Wave Polarization (Continued)

The right-handedness of the polarization implies that **E** rotates from the z axis towards the x axis. So, let the phase of $\tilde{E}_z$ be 0 and the phase of $\tilde{E}_x$ be $\delta = -\pi/2$. Then

$$\begin{aligned}\tilde{\mathbf{E}}(y) &= ae^{-jky}\hat{\mathbf{z}} + ae^{-j\pi/2}e^{-jky}\hat{\mathbf{x}} \\ &= a(-j\hat{\mathbf{x}} + \hat{\mathbf{z}})e^{-jky}. \qquad (71)\end{aligned}$$

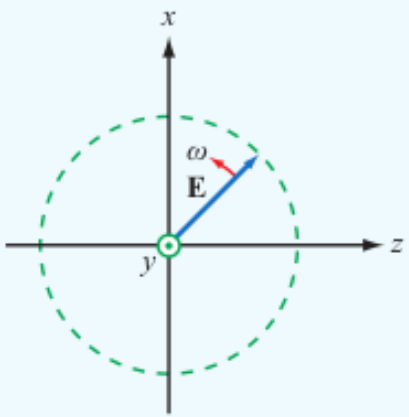

Figure 7-10 Right-hand circularly polarized wave of Example 7-2.

Therefore,

$$\begin{aligned}\mathbf{E}(y,t) &= \mathrm{Re}\left\{\tilde{\mathbf{E}}(y)e^{j\omega t}\right\} = \mathrm{Re}\left\{a(-j\hat{\mathbf{x}} + \hat{\mathbf{z}})e^{-jky}e^{j\omega t}\right\} \\ &= a\,\mathrm{Re}\left\{-je^{j(\omega t - ky)}\hat{\mathbf{x}} + e^{j(\omega t - ky)}\hat{\mathbf{z}}\right\} \\ &= (3\ \mathrm{mV/m})(\sin(\omega t - ky)\hat{\mathbf{x}} + \cos(\omega t - ky)\hat{\mathbf{z}}). \qquad (72)\end{aligned}$$

Wave Polarization (Continued)

Furthermore,

$$\tilde{\mathbf{H}}(y) = \frac{1}{\eta}\hat{\mathbf{y}} \times \tilde{\mathbf{E}} = \frac{1}{\eta}\begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ 0 & 1 & 0 \\ -aje^{-jky} & 0 & ae^{-jky} \end{vmatrix} = \frac{1}{\eta}\left(ae^{-jky}\hat{\mathbf{x}} + aje^{-jky}\hat{\mathbf{z}}\right)$$
$$= \frac{a}{\eta}(\hat{\mathbf{x}} + j\hat{\mathbf{z}})e^{-jky} \tag{73}$$

and

$$\mathbf{H}(y,t) = \mathrm{Re}\left\{\tilde{\mathbf{H}}(y)e^{j\omega t}\right\} = \mathrm{Re}\left\{\frac{a}{\eta}(\hat{\mathbf{x}} + j\hat{\mathbf{z}})e^{-jky}e^{j\omega t}\right\}$$
$$= \frac{a}{\eta}\,\mathrm{Re}\left\{e^{j(\omega t - ky)}\hat{\mathbf{x}} + je^{j(\omega t - ky)}\hat{\mathbf{z}}\right\}$$
$$= \frac{a}{\eta}(\cos(\omega t - ky)\hat{\mathbf{x}} - \sin(\omega t - ky)\hat{\mathbf{z}}). \tag{74}$$

Wave Polarization (Continued)

Here,

$$\frac{a}{\eta} = (3\ \text{mV/m})\left(\sqrt{\frac{\epsilon}{\mu}}\ 1/\Omega\right) = \left(3 \times 10^{-3}\ \text{V/m}\right)\left(\sqrt{\frac{4\epsilon_0}{\mu_0}}\ (1/\Omega)\right)$$
$$= \frac{6 \times 10^{-3}\ \text{V/m}}{\eta_0\ (\Omega)} \approx \frac{6 \times 10^{-3}}{120\pi}\ \text{A/m} = \frac{1}{20\pi}\ \text{mA/m} \tag{75}$$

so that

$$\mathbf{H}(y,t) = \left(\frac{1}{20\pi}\ \text{mA/m}\right)(\cos(\omega t - ky)\hat{\mathbf{x}} - \sin(\omega t - ky)\hat{\mathbf{z}}). \tag{76}$$

Since the y-components of $\mathbf{E}(y,t)$ and $\mathbf{H}(y,t)$ are 0, the electric and magnetic fields are both perpendicular to $\hat{\mathbf{y}}$.

Wave Polarization (Continued)

Also,

$$\begin{aligned}\mathbf{E}(y,t)\cdot\mathbf{H}(y,t) &= a(\sin(\omega t - ky)\hat{\mathbf{x}} + \cos(\omega t - ky)\hat{\mathbf{z}}) \\ &\qquad \cdot \frac{a}{\eta}(\cos(\omega t - ky)\hat{\mathbf{x}} - \sin(\omega t - ky)\hat{\mathbf{z}}) \\ &= \frac{a^2}{\eta}(\sin(\omega t - ky)\cos(\omega t - ky) \\ &\qquad - \cos(\omega t - ky)\sin(\omega t - ky)) \\ &= 0. \end{aligned} \tag{77}$$

This means that $\mathbf{E}(y,t) \perp \mathbf{H}(y,t)$ and the wave is TEM, as expected.

Wave Polarization (Continued)

Elliptical Polarization

When $\mathbf{E}(z,t)$ doesn't fit into one of the special cases described earlier, it traces an ellipse in any plane perpendicular to the direction of propagation at a fixed value of z and is **elliptically polarized**. The ellipse is specified by its **semi-major axis** a_ξ, its **semi-minor axis** a_η, and the **rotation angle**

$$-\pi/2 \leq \gamma \leq \pi/2 \tag{78}$$

the major axis makes with a reference direction, which is the $+x$ axis in this discussion. The **polarization is right- or left-handed**, following the same rules as for circular polarization.

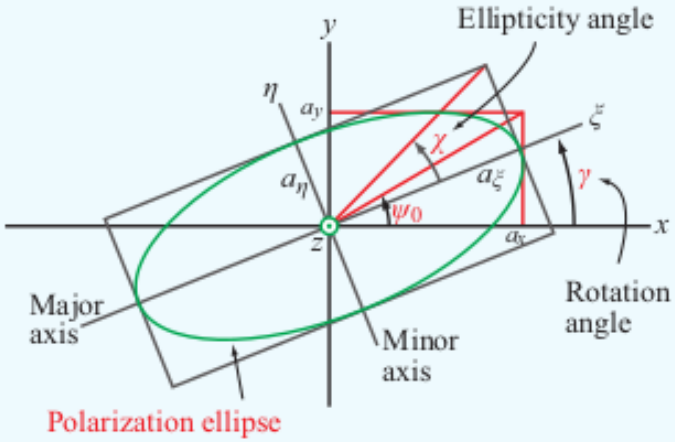

Figure 7-11 Polarization ellipse in the x–y plane, with the wave traveling in the z direction (out of the page).

Wave Polarization (Continued)

The coordinate ξ is defined along the major axis and the coordinate η is defined along the minor axis using the same origin and the same scales as the x-y-z coordinate system. (Note: The coordinate η is unrelated to the intrinsic impedance.)

The ellipse inscribes a $2a_\xi$ by $2a_\eta$ rectangle. The **ellipticity angle**

$$-\pi/4 \leq \chi \leq \pi/4 \tag{79}$$

between the $+\xi$ axis and direction of the first corner of the rectangle encountered in the direction that the electric field vector travels specifies the shape of the ellipse and the handedness of the polarization.

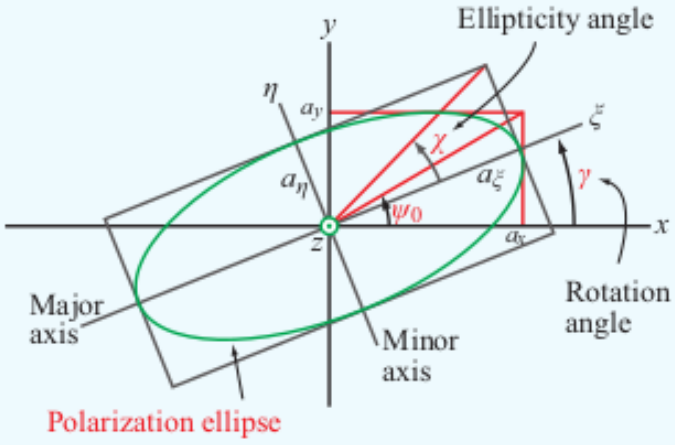

Figure 7-11 Polarization ellipse in the x–y plane, with the wave traveling in the z direction (out of the page).

Wave Polarization (Continued)

Here,

$$\tan\chi = \pm\frac{a_\eta}{a_\xi} = \pm\frac{1}{R}, \qquad (80)$$

where

$$R = \frac{a_\xi}{a_\eta} \qquad (81)$$

is the **axial ratio**. When $\tan\chi > 0$ (i.e. $\chi > 0$), the polarization is **left-handed** and, when $\tan\chi < 0$ (i.e. $\chi < 0$), the polarization is **right-handed**.

- When $R = 1$ (and $\chi = \pm\pi/4$), the semi-major and semi-minor axes are the same length and the polarization is **circular**.
- When $a_\eta = 0$, $\chi = 0$, $R = \infty$, and the polarization is **linear**.

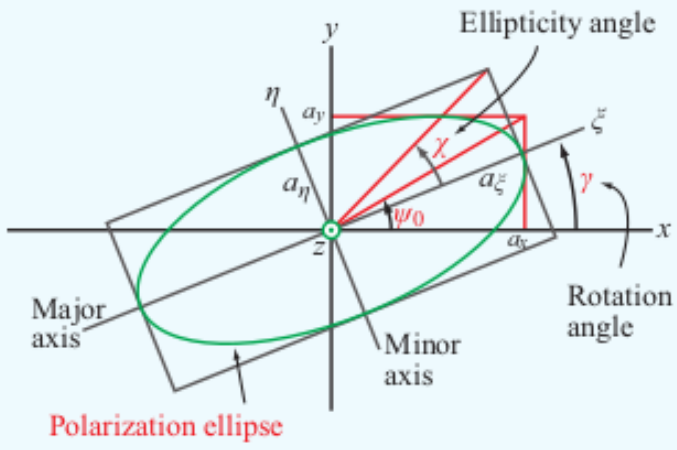

Figure 7-11 Polarization ellipse in the x–y plane, with the wave traveling in the z direction (out of the page).

Wave Polarization (Continued)

The rotation angle γ and the ellipticity angle χ can be determined using $a_x = |E_{x0}|$, $a_y = |E_{y0}|$, and the phase difference δ. Here,

$$\tan 2\gamma = (\tan 2\psi_0)\cos\delta \tag{82}$$

$$\sin 2\chi = (\sin 2\psi_0)\sin\delta, \tag{83}$$

where

$$\psi_0 = \tan^{-1}\left(\frac{a_y}{a_x}\right) \tag{84}$$

is the **auxiliary angle** with

$$0 \le \psi_0 \le \pi/2. \tag{85}$$

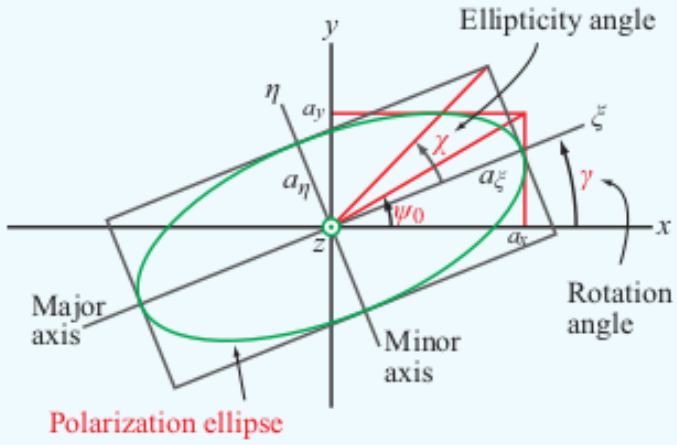

Figure 7-11 Polarization ellipse in the x–y plane, with the wave traveling in the z direction (out of the page).

The equation for γ has an ambiguity that is resolved by requiring γ to have the same sign as $\cos\delta$.

Wave Polarization (Continued)

Example: Determining Polarization State

Find the polarization state of a uniform plane wave with

$$\mathbf{E}(z,t) = 3\cos\left(\omega t - kz + \frac{\pi}{6}\right)\hat{\mathbf{x}} - 4\sin\left(\omega t - kz + \frac{\pi}{4}\right)\hat{\mathbf{y}}. \tag{86}$$

The first step is to find the phasor $\tilde{\mathbf{E}}(z)$ of $\mathbf{E}(z,t)$. Since the real part of a phasor must be a cosine, this means converting the sine function into a cosine. Here,

$$\sin\left(\omega t - kz + \frac{\pi}{4}\right) = \cos\left(\omega t - kz + \frac{\pi}{4} - \frac{\pi}{2}\right) = \cos\left(\omega t - kz - \frac{\pi}{4}\right) \tag{87}$$

and

$$\mathbf{E}(z,t) = 3\cos\left(\omega t - kz + \frac{\pi}{6}\right)\hat{\mathbf{x}} - 4\cos\left(\omega t - kz - \frac{\pi}{4}\right)\hat{\mathbf{y}}. \tag{88}$$

Wave Polarization (Continued)

So,

$$\tilde{\mathbf{E}}(z) = 3e^{j(\pi/6-kz)}\hat{\mathbf{x}} - 4e^{-j(\pi/4+kz)}\hat{\mathbf{y}} = 3e^{j\pi/6}e^{-jkz}\hat{\mathbf{x}} - 4e^{-j\pi/4}e^{-jkz}\hat{\mathbf{y}}$$
$$= \left(3e^{j\pi/6}\hat{\mathbf{x}} - 4e^{-j\pi/4}\hat{\mathbf{y}}\right)e^{-jkz}. \tag{89}$$

To match the form

$$\tilde{\mathbf{E}}(z) = \left(a_x\hat{\mathbf{x}} + a_y e^{j\delta}\hat{\mathbf{y}}\right)e^{-jkz}, \tag{90}$$

the coefficients need to be positive. Since $e^{j\pi} = -1$,

$$\tilde{\mathbf{E}}(z) = \left(3e^{j\pi/6}\hat{\mathbf{x}} + 4e^{j\pi}e^{-j\pi/4}\hat{\mathbf{y}}\right)e^{-jkz} = \left(3e^{j\pi/6}\hat{\mathbf{x}} + 4e^{j3\pi/4}\hat{\mathbf{y}}\right)e^{-jkz}. \tag{91}$$

The phase difference between the $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ components of the phasor is

$$\delta = \delta_y - \delta_x = \frac{3\pi}{4} - \frac{\pi}{6} = \frac{7\pi}{12}. \tag{92}$$

Wave Polarization (Continued)

Setting the phase of the $\hat{\mathbf{x}}$ component to 0 and the phase of the $\hat{\mathbf{y}}$ component to δ,

$$\tilde{\mathbf{E}}(z) = \left(3\hat{\mathbf{x}} + 4e^{j7\pi/12}\hat{\mathbf{y}}\right) e^{-jkz}. \tag{93}$$

Here, $a_x = 3$, $a_y = 4$, and $\delta = 7\pi/12$. What type of polarization does the wave have? We already know that its elliptical since $a_y \neq a_x$ (This excludes circular polarization.) and $\delta \neq 0, \pi$ (This excludes linear polarization.).

The auxiliary angle is

$$\psi_0 = \tan^{-1}\left(\frac{a_y}{a_x}\right) = \tan^{-1}\left(\frac{4}{3}\right) = 0.93 \text{ rad} = 0.30\,\pi \text{ rad} = 53.1^\circ. \tag{94}$$

Furthermore,

$$\tan 2\gamma = (\tan 2\psi_0)\cos\delta = \tan(106.2^\circ)\cos(105^\circ) = 0.89 \tag{95}$$

and

$$\gamma = 0.36 \text{ rad } (= 20.8^\circ) \text{ or } -1.21 \text{ rad } (= -69.2^\circ). \tag{96}$$

Wave Polarization (Continued)

Since γ and $\cos\delta$ must have the same sign and

$$\cos\delta = \cos\left(\frac{7\pi}{12}\right) = -0.26, \tag{97}$$

$$\gamma = -1.21 \text{ rad } (= -69.2^\circ). \tag{98}$$

Finally,

$$\sin(2\chi) = (\sin 2\psi_0)\sin\delta = 0.93 \implies \chi = 0.59 \text{ rad } (= 34.0^\circ). \tag{99}$$

Since $\chi > 0$, the polarization is left-handed. Also,

$$R = \frac{1}{\tan\chi} = 1.48. \tag{100}$$

Homework

Read Chapter 7.

Homework 1 is posted and is due Tuesday by the beginning of class.